Islamic University of Technology

Computer Networks Lab

CSE 4512

Lab 1 Report

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1. Task Description

In Islamic University of Technology (IUT), each computer lab has its own network. There are five labs, each with two PCs. The lab network follows a pattern using 192.id for the first two portions of the network address, where *id* is the last two digits of your student ID. Your task is to design an interconnection such that any PC in Lab x can reach any PC in Lab y.

As IUT is located in Board Bazar, frequent load shedding can cause some network devices to stop functioning temporarily. This must be considered when designing the network to ensure uninterrupted communication.

2. Solution Approach

The network design for the five computer labs at IUT was implemented as follows:

1. **Building the Topology:**
    - Each PC is connected to its lab's access switch, forming the local network of the lab.
    - Each access switch is connected to the lab's access router, creating a stable LAN environment for the lab.
    - Some lab access routers are interconnected among themselves and also connected to a central switch (except one), forming a hybrid topology.
    - The hybrid topology combines a partial mesh and a centralized star structure, ensuring multiple paths between labs to handle load-shedding issues.
2. **Configuring the PCs:**
    - Each PC was assigned a unique IP address within the 192.22.10.x addressing scheme.

- The default gateway for each PC was set to its lab's access router.

3. **Configuring the Routers:**
   - Each access router was assigned an IP address for its lab network following the 192.22.x.y scheme.
   - Router-to-router links were configured and enabled to ensure inter-lab connectivity.

4. **Verifying Connectivity (Ping Tests):**
   - Connectivity was verified by pinging from one PC to another within the same lab.
   - Global connectivity was tested by pinging from one PC in a lab to PCs in other labs.

5. **Solving the Load Shedding Problem:**
   - The hybrid topology ensures that multiple paths exist between labs.
   - No single direct connection is a critical point of failure, allowing uninterrupted communication even if a router or switch loses power.
   - The central switch serves as a secondary backbone, increasing survivability during power outages.

This overall design guarantees that every PC in Lab x can reach every PC in Lab y, fulfilling the connectivity requirement and ensuring resilience against load-shedding. The hybrid structure creates a robust network suitable for the local conditions of IUT.

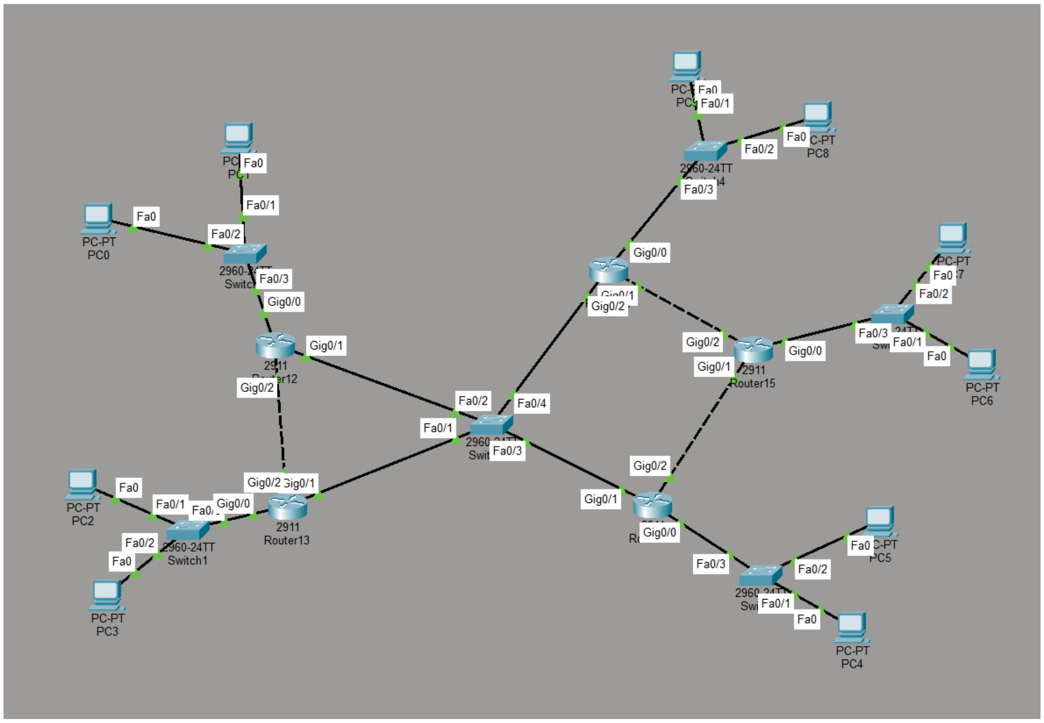

Figure 1: Network Structure